OB+RCT
CATE/Overlap/High dimension/Matching

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Overview

- why combine rct and ob (poor overlap in ob)
- review of overlap assumption in causal inference (IPW, PLM, R-learner, CRF)
- how to measure overlapness and balance
- overlap/balance intuited method
- question: how can rct be better combined with ob (in a 'better overlapness/balance therefore better statistical properties' sense)?

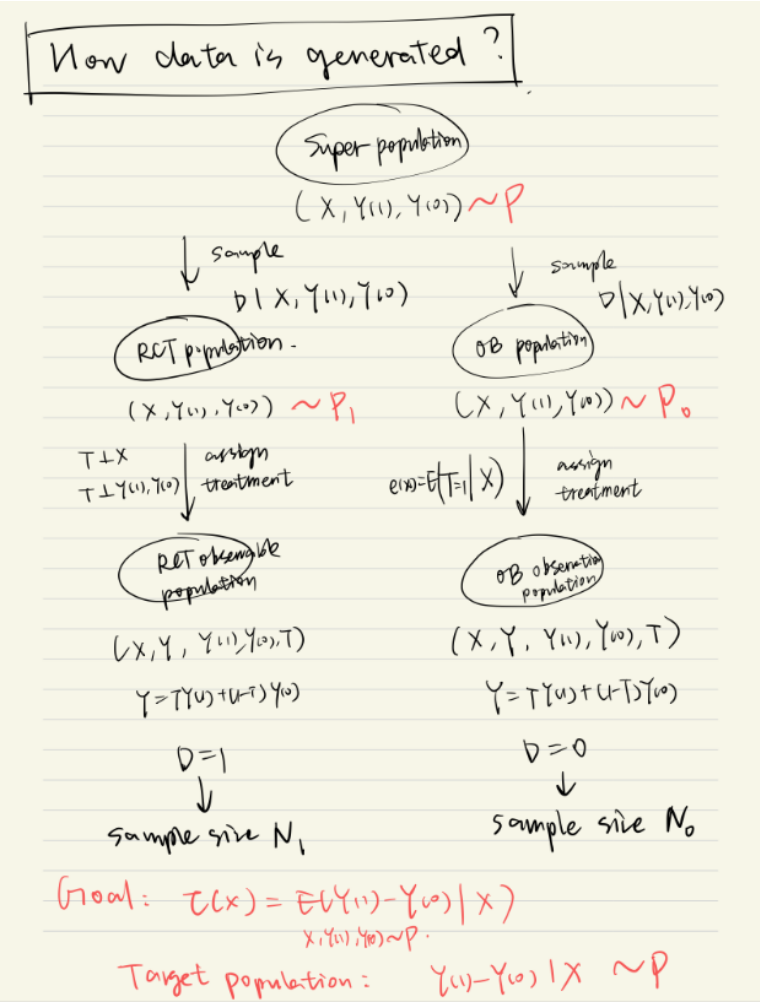
why combine rct and ob, and super-population framework

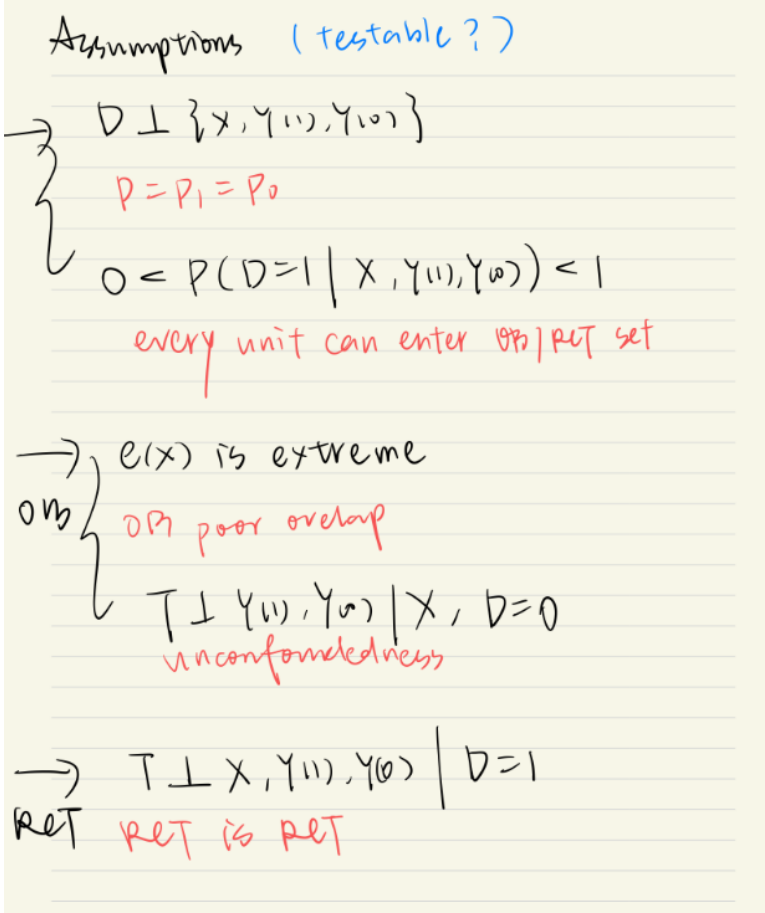
combine rct and ob - why?

- RCT+EC
  - RCT so small
  - RCT may only cover some specific patient(X support region)
- OB+RCT
  - OB with poor overlap(in what sense?) our case
  - OB with unmeasured confounding

combine rct and ob - a data generating view

In super population framework, both RCT and OB are samples from a stochastic distribution. (not finite population and design-based experiment framework) (Liu, L. Thomas, et al. 2025) introduce a double-indexed notation.





review of the role of overlap assumption

OB - why overlap important

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)
- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for statistical analysis in many papers. (e.g. Causal RF)

Overlap in IPW

(Wager 2024)

- oracle IPW estimator is asymptotically normal

$$\sqrt{n}(\hat{\tau}^*_{ipw} - \tau) \sim N(0, V^*_{IPW})$$

where the asymptotical variance

$$V_{IPW^*} = \mathrm{Var}\left[\tau(X_i)\right] + \mathbb{E}\left[\frac{\left(\mu_{(0)}(X_i) + (1 - e(X_i))\tau(X_i)\right)^2}{e(X_i)(1 - e(X_i))}\right] + \mathbb{E}\left[\frac{\sigma^2_{(1)}(X_i)}{e(X_i)} + \frac{\sigma^2_{(0)}(X_i)}{1 - e(X_i)}\right]. \quad (1)$$

- all moments need to be well-defined and finite
  - if potential outcomes are uniformly bounded, it means we need $E(\frac{1}{e(X)(1-e(X)})) < \infty$
  - if the potential outcomes have finite second moments, it means we need bounded propensity, i.e. strict overlap

Overlap in PLM

Consider Partial linear model

$$E(Y \mid X, Z) = \beta^T Z + \theta(X)$$

- (Robinson 1988) (generalized FWL theorem) regression-of-residuals estimator is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if
  - $\Phi = E[(Z - E(Z|X))(Z - E(Z|X))^\top]$ is positive definite.
    - $\Phi^{-1} = \frac{1}{E(e(x)(1-e(x))} = \frac{1}{E(Var(Z|X))}$ (convergence rate & overlap)
  - kernel estimators for first step, other ..

Overlap in R-learner

Considering HTE, now the PLM model becomes:

$$E(Y \mid X, Z) = \beta(X)Z + \theta(X)$$

- (Nie and Wager 2021) equivalently

$$\tau^*(\cdot) = \arg\min_{\tau} \mathbb{E}\left[\left(Y - m^*(X) - (W - e^*(X))\tau(X)\right)^2\right]$$

  therefore estimator(minimize R-loss, cross fitting):

$$\hat{\tau} = \arg\min_{\tau} \left(\hat{L}_n(\tau) + \Lambda_n(\tau)\right)$$

  - penalized kernel R-learner is near oracle(depends on the local robustness property of R-loss) if
    - strict overlap: turning regret bound into MSE bound

$$(1-\eta)^{-2}R(\tau) < E[\{\tau(X_i) - \tau^*(X_i)\}^2] < \eta^{-2}R(\tau)$$

    - RKHS, good estimate of nuisance functions, kernel, ...
  - when $e(X) \approx 0$, R-learner roughly equivalent to X-learner(?)

Overlap in CRF - high-dimensional/heterogeneity

(Wager and Athey 2018)

- Bias of estimation

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular

more on overlapness and balance

OB - overlap

- why poor overlap?
  - highly but not totally deterministic assignment mechanism(e.g. medical interventions often based on guidelines)
  - side question: is the assignment mechanism good (propensity is high when effect is high)? (e.g. is the guideline optimal for patient?) Would that affect how we improve the current guideline? make better estimation by ob and rct)
- when does poor overlap matter?
  - side ques above
  - unbalance of confounders[?] has more influence
- poor overlap, then?
  - need more data, conduct rct
  - how much rct samples do we need? better ways to get rct?
  - how to use these new rct data?

OB - overlap & balance

Define P_1 by $\frac{dP_1}{dP} = \frac{e(X)}{E_p(e(X))}$ and P_0 by $\frac{dP_0}{dP} = \frac{1-e(X)}{E_p(1-e(X))}$, which are the distribution of covariates in treatment and control arm. Let $P(Z=1) = \pi$ be a constant $\pi \in (0,1)$.
By Bayes' theorem, we have density ratio(a measure of balance) with propensity ratio(a measure of overlap) as

$$\frac{dP_1}{dP_0} = \frac{P(X|Z=1)}{P(X|Z=0)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{P(Z=0)}{P(Z=1)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{1-\pi}{\pi}$$

and the mean discrepancy(a measure of balance) is

$$E(X|Z=1) - E(X|Z=0) = E_{P_1}(X) - E_{P_0}(X)$$
$$= E_P(X(\frac{dP_1}{dP} - \frac{dP_0}{dP})) = E_P(X(\frac{e(X)-\pi}{\pi(1-\pi)}))$$

Also can derive other f-divergence form of distance to measure the difference of the two distributions.

$$D_f(Q_1 \| Q_0) := E_{Q_0}[f(\frac{dQ_1}{dQ_0})]$$

OB - strict overlap & balance in high-dimensional setting

high-dimensional: (1) many covariates; (2) complex interactions among covariates in outcome model or propensity model

- Strict overlap with η implies bounded density ratio.

$$b_{min} =: \frac{1-\pi}{\pi}\frac{\eta}{1-\eta} \leq \frac{dP_1}{dP_0} \leq \frac{1-\pi}{\pi}\frac{1-\eta}{\eta} := b_{max}$$

- Strict overlap restricts global discrepancies between P_0 and P_1([support same?]), and these restrictions become more restrictive as the dimension of covariates grows large(D'Amour et al. 2021)[1]

$$D_f(P_1 \| P_0) \leq B_f(b_{min}, b_{max})$$

$$D_f(P_0 \| P_1) \leq B_f(b_{max}^{-1}, b_{min}^{-1})$$

- From data-generating view, if we have high-interaction confounding, then poor high-dimension overlap [?]

[1] where the upper bound B_f is actually free of p, while the left side may not. So strict overlap is very 'strict' in high-dimensional setting.

OB - overlap easy to satisfy? an example

Look into the classical setting:

$$e(X) = exp(W_e),\ X \sim N_p(\mu, \Sigma),\ W_e = \beta_0 + \sum_{j=1}^{p} \beta_j X_j \sim N(\beta_0 + \beta^T \mu, \beta^T \Sigma \beta)$$

Does strict overlap hold? for a constant $\eta \in (0, 1)$.
What would influence the divergence $D_f(P_1 || P_0)$?

OB - measure of population overlapness[2]

Overlapness can be a overall concept, or a local concept.

- how strong tails of $e(x)$
- $min(\eta)$ that satisfy $\eta < e(X) < 1 - \eta$ (intuited by strict overlap)
- $E(e(x)(1 - e(x))$(a bounded measure) ?
- (Liu and Li 2025) Let $f_z(u) = Pr(e(X) = u \mid Z = z)$, the Bhattacharyya coefficient

$$\phi \equiv \int_0^1 \sqrt{f_1(u) f_0(u)}\, \mathrm{d}u.$$

  - assume propensity function is canonical a logit-normal distribution(by Lyapunov CLT), which can be approximated by Beta distribution
  - (r, ϕ) is sufficient to define this beta dist
  - given r, ϕ is monotonically increasing in both a and b.
- (Kim et al. 2025) Let $\Sigma_z = \mathbb{E}[\phi(x) \otimes \phi(x)\mathbf{1}(Z = z)]$, define weak overlap[?]

$$\Sigma_0 \preceq R\left(\Sigma_1 + \frac{\xi}{n}\mathbf{I}\right) \quad \text{and} \quad \Sigma_1 \preceq R\left(\Sigma_0 + \frac{\xi}{n}\mathbf{I}\right)$$

[2] for sample analogue, just use the min value or impute $\hat{e}(x)$

OB - measure of sample balance

Balance is an overall concept(no local version), and we need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[3]); mean-matching/pair-matching (Rubin 1973a)
- Mahalanobis distance[4] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[3] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[4] MD does not perform well when covariates have nonellipsoidal distributions?

review of overlap/balance-intuited methods

overlap/balance-intuited methods

- (Li, L. E. Thomas, and Li 2019) [ATE] Overlap Weighting
- ? (Crump et al. 2009) [ATE] Characterize(by ps score) optimal subsamples for which the average treatment effect can be estimated most precisely.

we also consider the problem without imposing this restriction. Formally, we choose an estimand $\tau_{S,\mathbb{A}}$ by choosing the set $\mathbb{A} \subset \mathbb{X}$ that minimizes

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{\{E(1_{X\in\mathbb{A}})\}^2} E\left[1_{X\in\mathbb{A}}\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)}\right\}\right].$$

Defining $q(\mathbb{A}) = \text{pr}(X \in \mathbb{A}) = E(1_{X\in\mathbb{A}})$, we can write the objective function as

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{q(\mathbb{A})} E\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)} \middle| X \in \mathbb{A}\right\}.$$

We seek $\mathbb{A} = \mathbb{A}^*$, which minimizes $V_{\mathbb{A}}^{\text{eff}}$ among all closed subsets $\mathbb{A} \subset \mathbb{X}$.

overlap/balance-intuited methods - more on CRF

(Wager and Athey 2018)

- Bias of estimation

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

- CRF is a data-driven way to determine which nearby observations receive more weight
- Tree beats NN under settings of low-dimensional signal structure. When poor overlap in a high-order-interaction-terms sense, hard to grow a tree?
- More questions
  - is strict overlap plausible for tree? how small can η be?
  - how does sample balance influence the structure(depth, t/c sample ratio in leaves, overall balance of tree(?)[5]), and then the performance of the tree?
  - could match/balance based weighting be used in CRF? (e.g. a balance-aware way of sample weighting/splitting[6])

[5] look at the assumptions

[6] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

OB+RCT, matching weighting, simulation, etc

OB - matching

- (Rubin 1973b) [OB, ATT, one-dimension X, homogeneous/parallel effect]. Use percent reduction in bias as a metric. Matching and regression can reduce bias in observational studies.
- (Diamond and Sekhon 2013) The GenMatch algorithm searches a range of distance metrics to find the particular measure that optimizes post-matching covariate balance. iteratively checking the specification of the propensity score model.

OB - matching and weighting

https://www2.stat.duke.edu/~fl35/teaching/640/Chap3.4_observational_weighting.pdf

Connection to Matching and Regression

- Matching: link "similar" cases in two samples, discard unmatched cases (bottom-up approach)
- Weighting: apply weights to entire samples, designed to create global balance (top-down approach)
- Intrinsic connection: Overlap weighting approaches many-to-many matching as the propensity score model becomes increasingly complex
- Why? The limit of a PS model is a saturated model with a fixed effect (dummy variable) for each design point (or small neighborhood, for continuous variables)

Connection to Matching and Regression

- Let the sample count for x_i in group $z = 0, 1$ be N_{zi}, the estimated PS from the saturated model is $\hat{e}(x_i) = N_{1i}/(N_{0i} + N_{1i})$
- The overlap weight for trt and con units is $N_{0i}/(N_{0i} + N_{1i})$ and $N_{1i}/(N_{0i} + N_{1i})$, respectively
- In many-to-many matching: at each design point x, each trt unit is matched to the N_{0i} controls at the same x, each control to N_{1i} trt units – exactly the same as the overlap weighting
- The total overlap weight for each design point x and hence of $\hat{\tau}(x_i)$ is $N_{0i}N_{1i}/(N_{0i} + N_{1i})$
- This is exactly the precision weight attached to $\bar{y}_{1i} - \bar{y}_{0i}$ in the fixed-effects for each design point OLS outcome model $Y_{zi} = \alpha_i + z\tau + \epsilon_{zi}$

combine rct and ob

- simple-but-possibly-wrong method: just combine the two source dataset, and throw it into model
- naive method: separately estimate and average the results.
  - just a linear combination of two estimators, easy to analyze its properties
  - involve tuning(more smart way of tuning?), not end-to-end

Simulations settings

See `https://oftrongs.github.io/Causal/simulation_settings.pdf` for summary of simulation settings in some papers.

Under what kind of settings would our methods be better than other?

N obs data, M rct data; d covariates, within which c are confounders and $d - c$ are prognsotic covariates(assume no instrumental variables)

- $e(x)$, overlap, well-specified or not
- M/N, relative information size of rct sample
- $\mu_0(x), \mu_1(x)$, the response surfaces and CATE $\tau(x)$
- c/d, how many irrelevant covariates
- high-dimensional or not (d/N or c/N, complexity of models)

Simulations settings - how to simulate X

Types of X: xx
(D'Amour et al. 2021) Covariate sequence X is a stochastic process $(X_{(k)})_{(k>0)}$, for any single problem, the investigator selects a finite set of covariates $X_{1:p}$ from the finite pool of covariates $(X_{(k)})_{(k>0)}$.

Simulations settings - linearity of outcome model

What comes along with linear model assumption? What if this is not true?

If the true outcome model is linear, $\mu_{(w)}(x) = x\beta(w)$, then

- the difference of them, i.e. CATE, is linear in X
- fitting one model for each z is equivalent to adding an interaction term Xz
- achieving sample balance[7] just involves balancing the X_i
- If the response surface is nonlinear, all estimates are general biased even if $\bar{x}_1 = \bar{x}_2$. (Rubin 1973b) [ATT, one-dimension]

[7] $\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i\phi_j(X_i)}{e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)\phi_j(X_i)}{1-e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\phi_j(X_i)$

Some thoughts and questions

- why does overlap influence the performance of estimators in math[8]
- overlap/balance aware adaptive algorithm?
- What is target population? After matching/weighting, would that actually change? (can visualize X)
- After matching/weighting, is the balance better?
- metrics of CATE estimation for simulation? [9]
- poor overlap in high-order-interaction-terms sense? hard to grow a tree? [how much overlap, i.e. how strong tails of $e(x)$]
- (matching-based) data-augmenting then modeling v.s. sample-weighting in DL/Tree models. Would it be different if we double one sampleor weight it by two in model?

[8] 'If the imbalance of the covariates between the two groups is large, the model-based results heavily relies on extrapolation in the region with little overlap, which is sensitive to the model specification assumption.'

[9] (1) $\frac{1}{n}\sum_{i=1}^{n}(\hat{\tau}_i - \tau_i)^2$; (2) $\frac{1}{n}\sum_{i=1}^{n}\hat{\tau}(x_i) - \frac{1}{n}\sum_{i=1}^{n}\tau(x_i)$; (3) $E(\hat{\beta} - \beta)^2$

Summary(before)

- Goal: better estimate CATE
  - data: large-scale Post-OB data and small-scale RCT data
  - problem: poor overlap and severe unbalance
- Method: matching-based
  - data augmentation
  - ITE-imputing loss
- Data source: simulation and real-world data
- Further
  - when would the methods work better(the choice of caliper for matching, the size of ob&rct, the degree of non-overlap, the use of previous algorithm or not, etc)
  - an end-to-end rather than step-wise algorithm
  - multi-treatment

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